Lab 01

Analog and Mixed Signal IC Design

Introduction to Cadence

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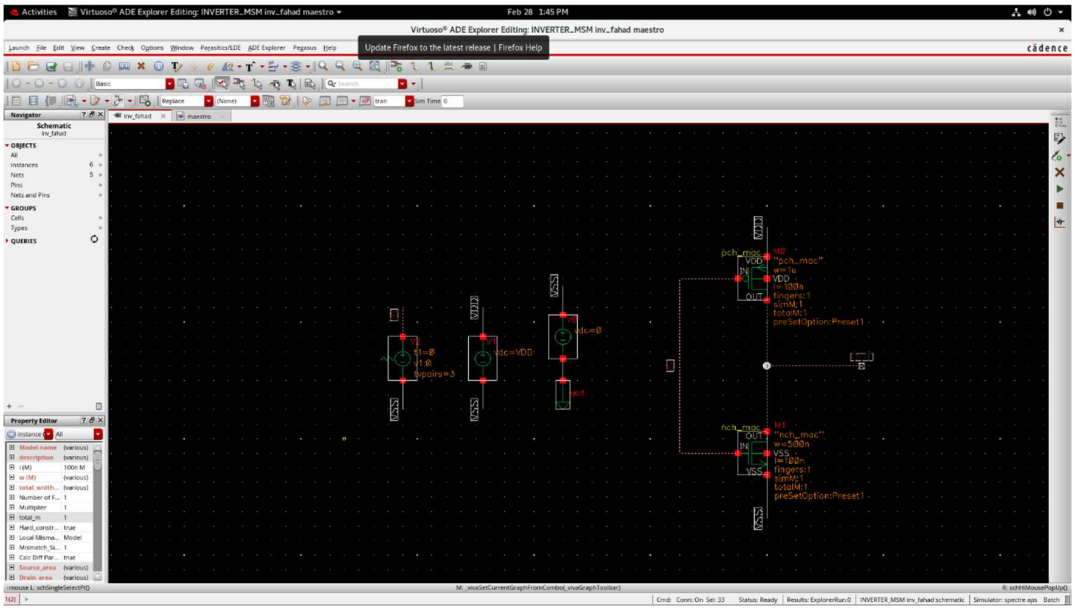
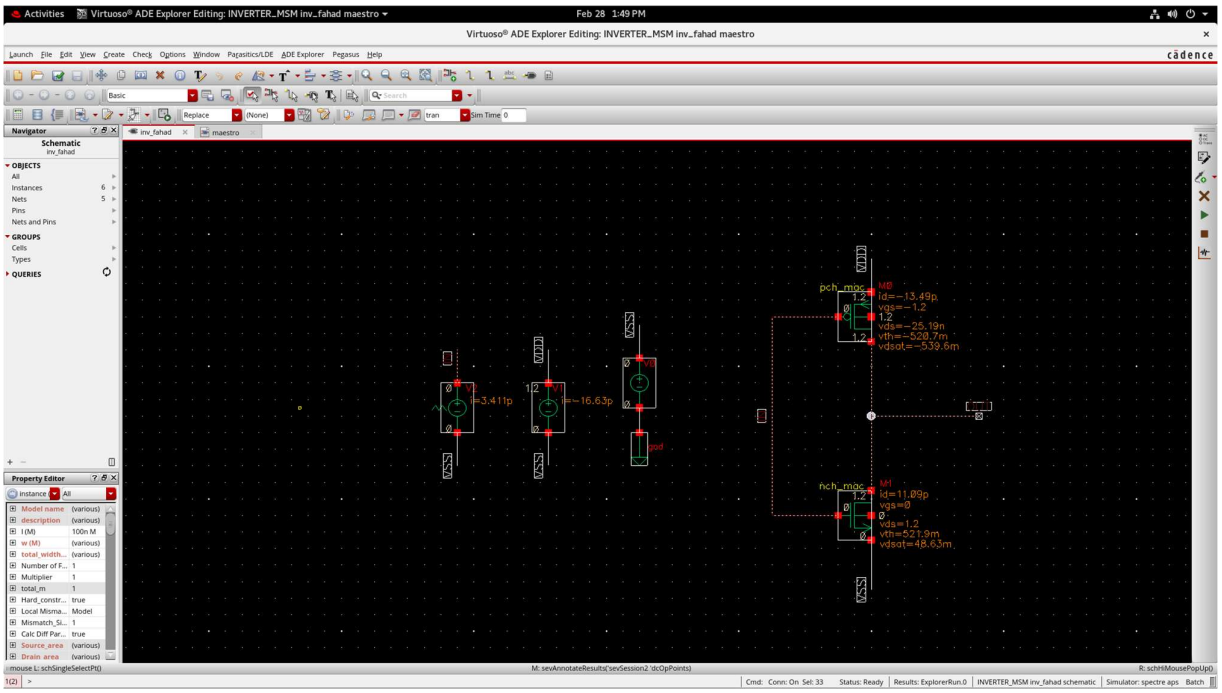
Pre Lab/Tutorial Task

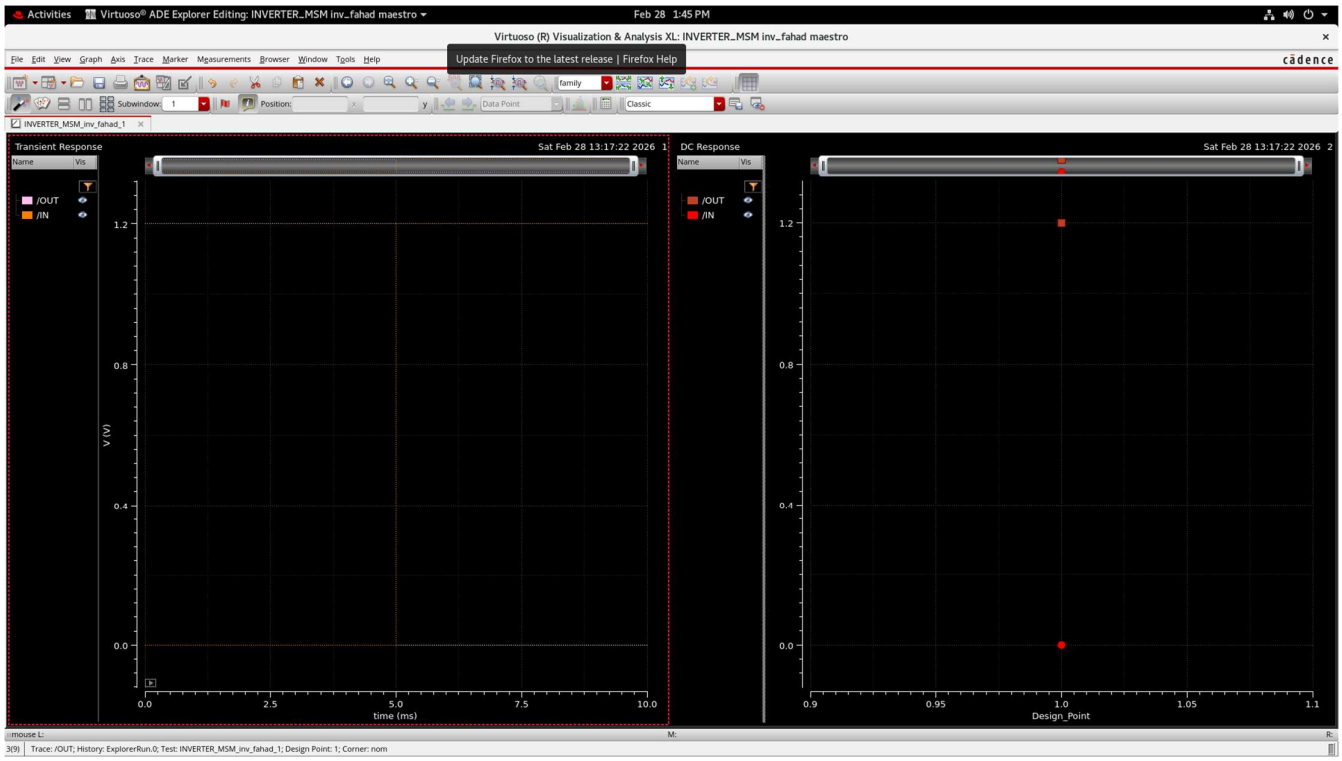
- Circuit



- DC Node voltages and Operating Points



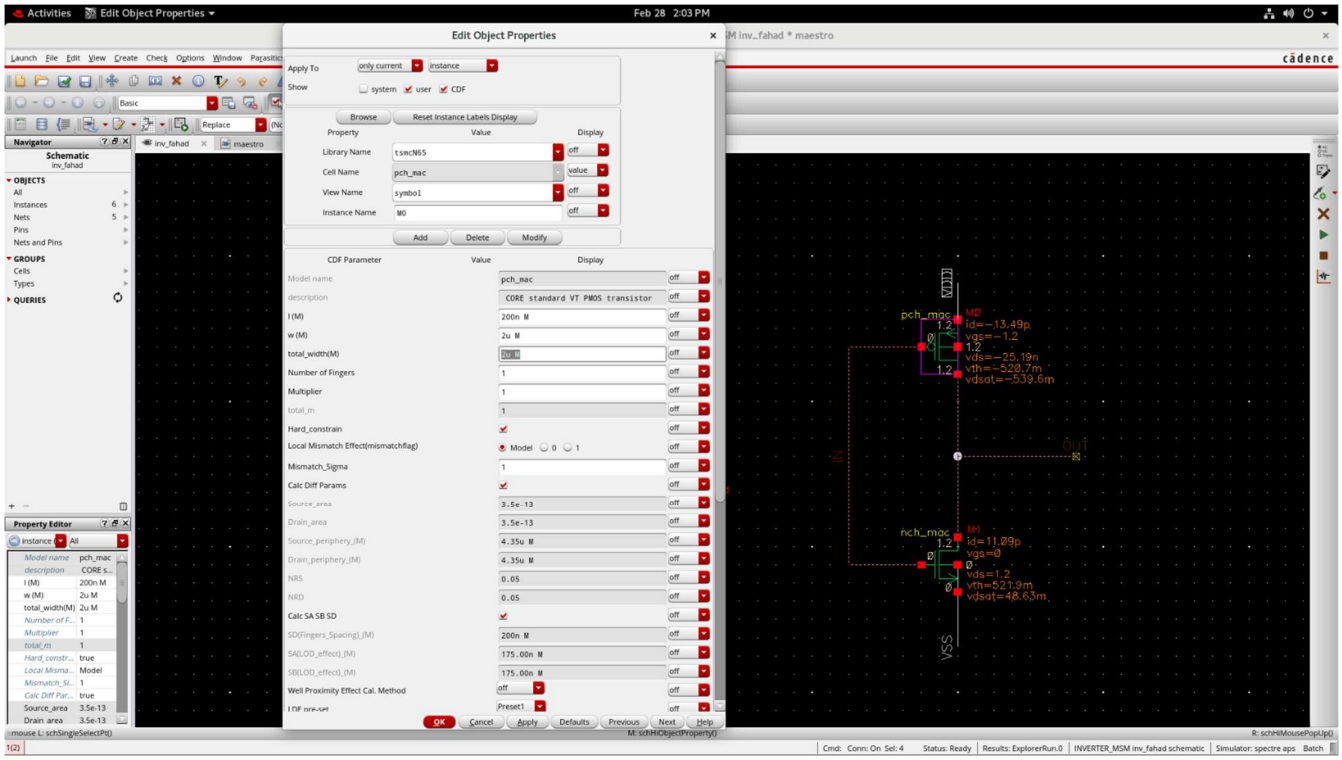
- Transient and Small Signal Analysis



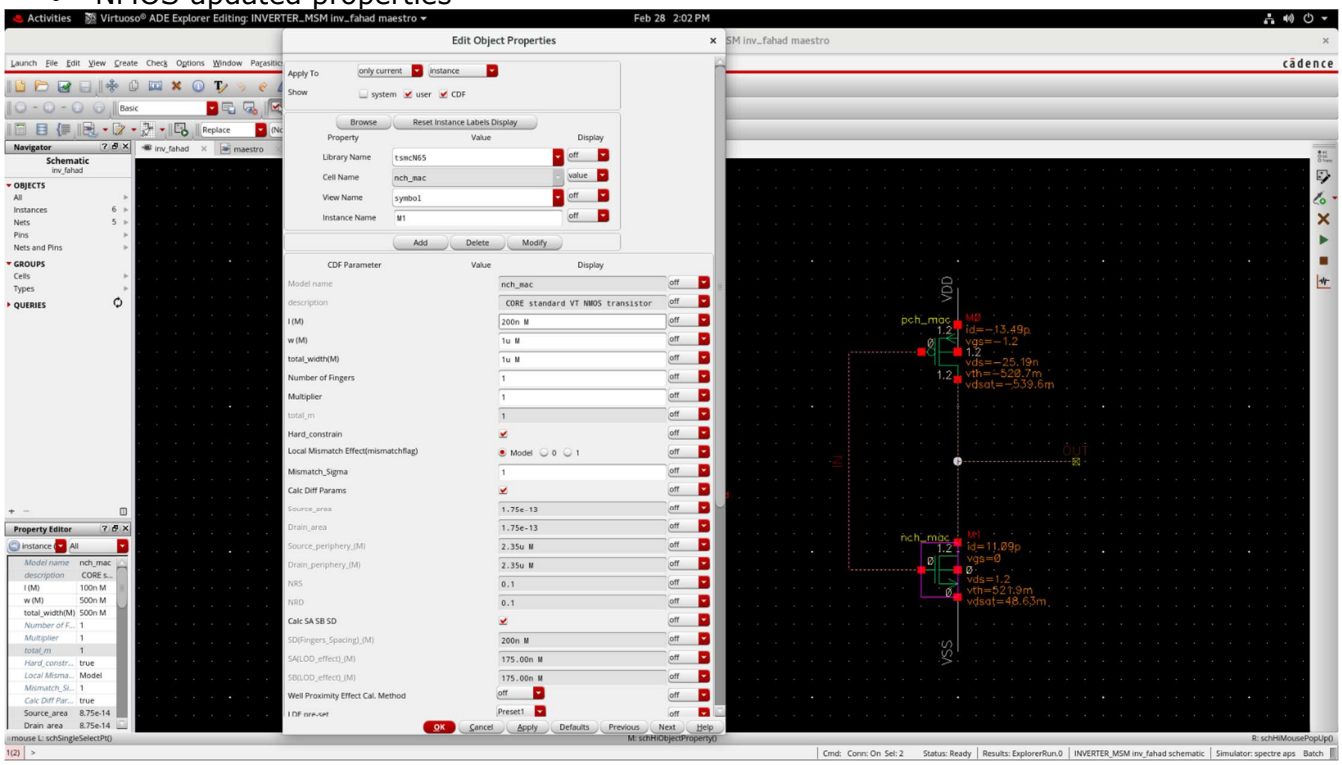
Exercise 1:

Repeat the exercises above with highest sizes of PMOS and NMOS transistors, observe and generate the current graphs and compare the results with the previous task.

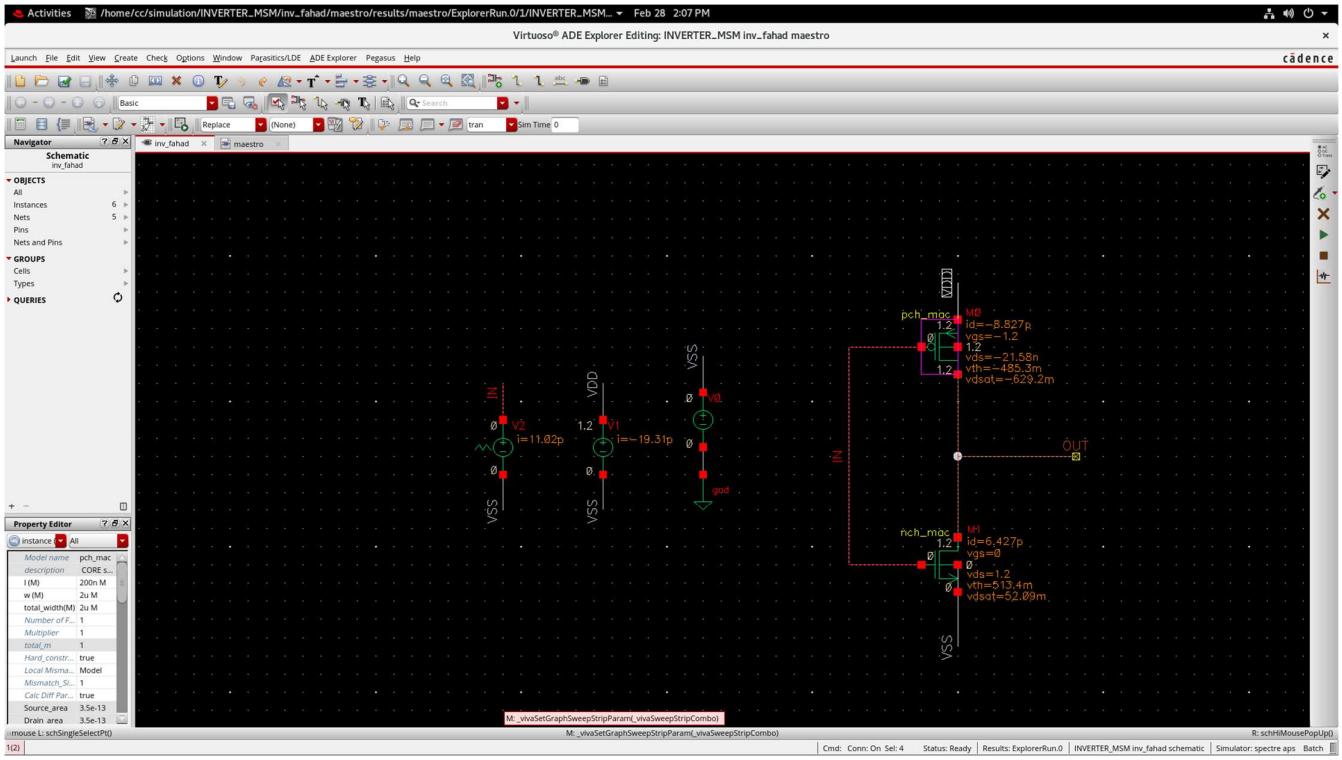
- PMOS updated properties



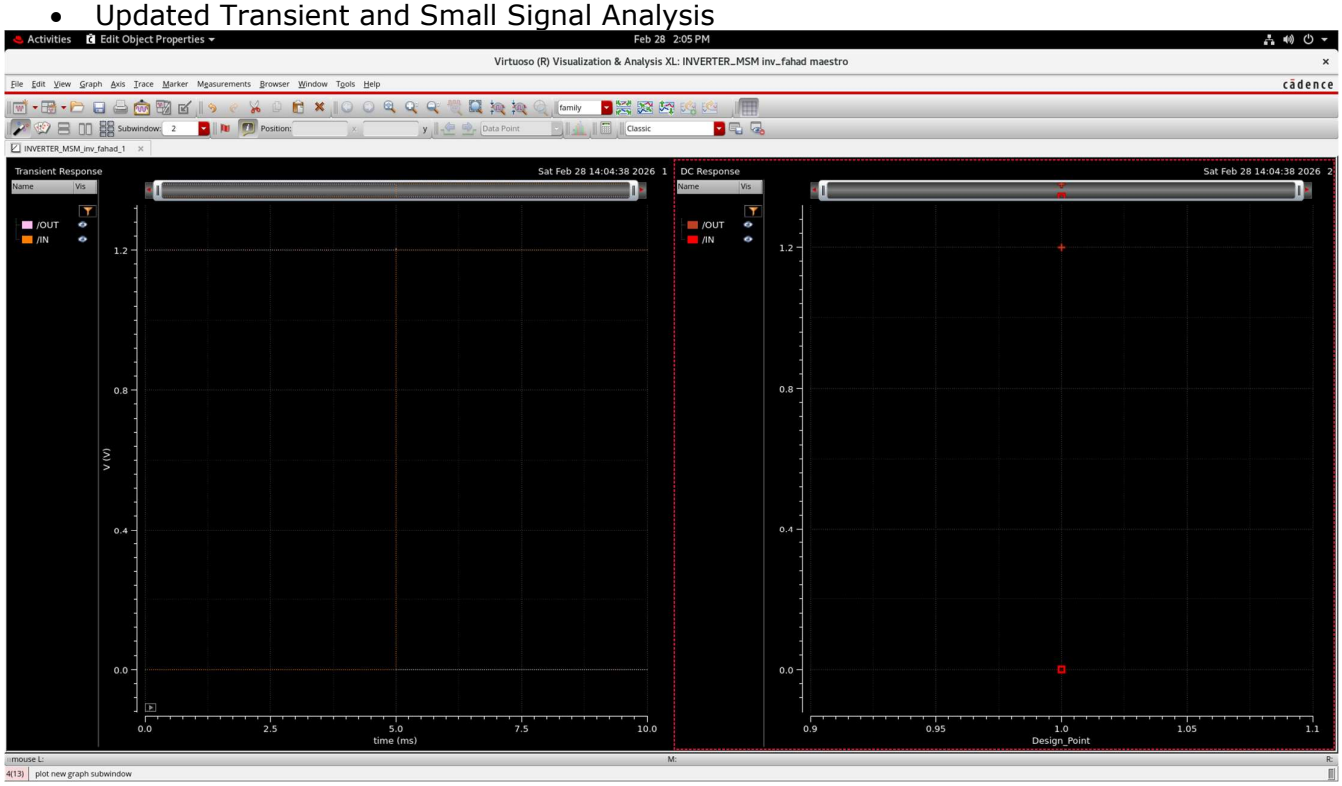
- NMOS updated properties



- Updated DC voltages and operating points

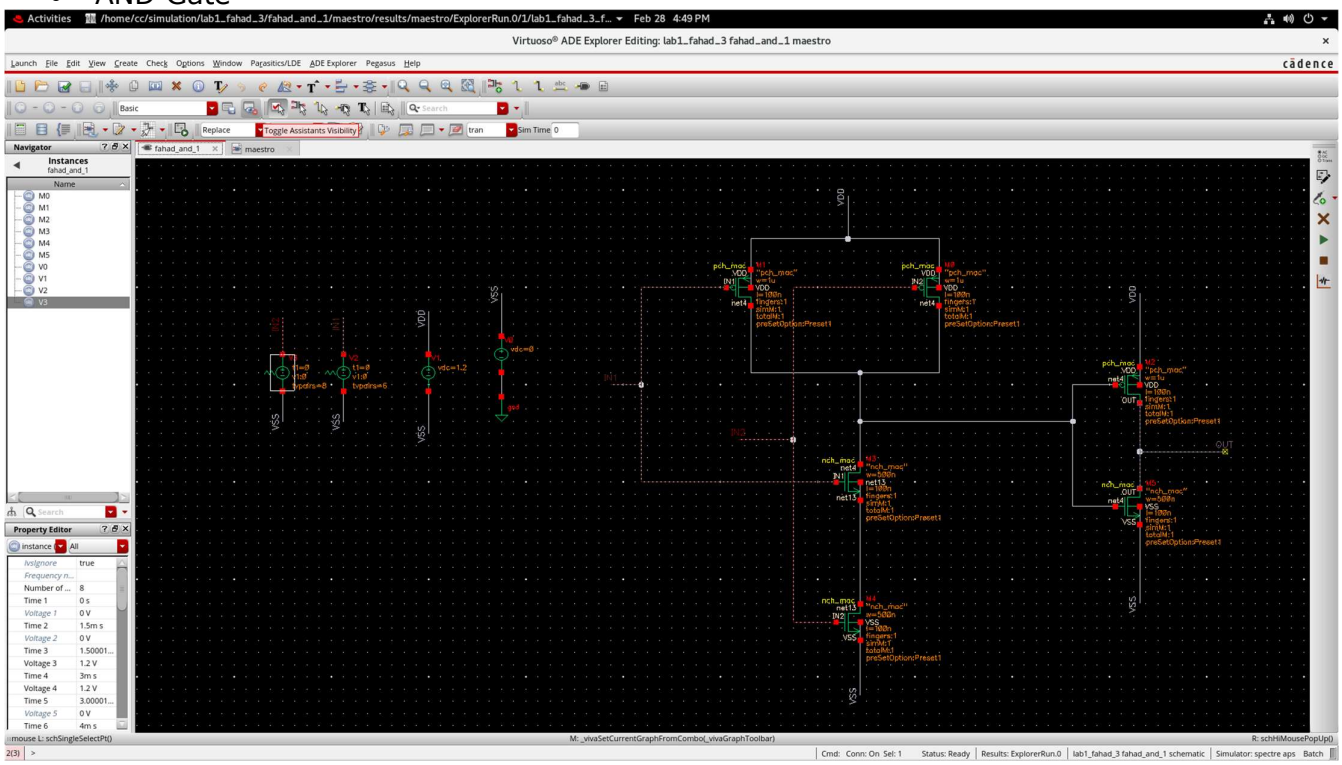


- Updated Transient and Small Signal Analysis

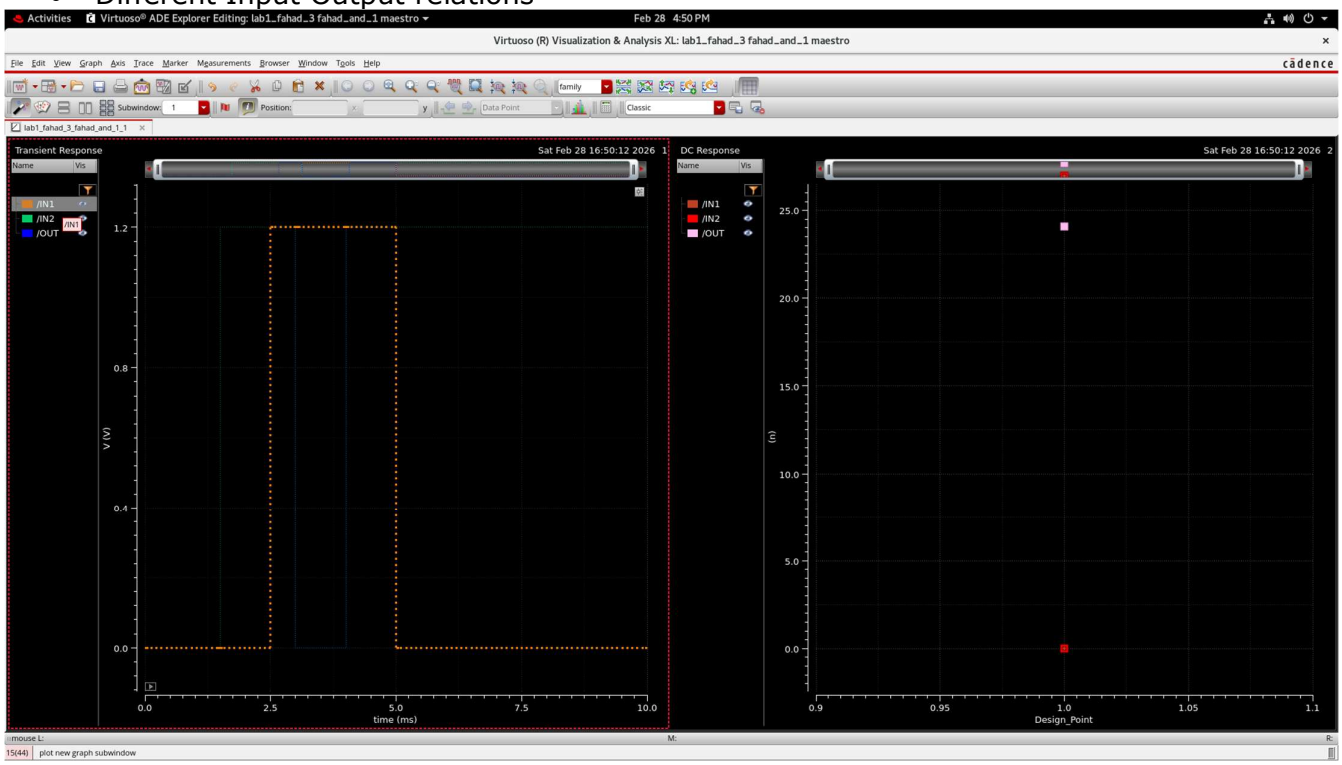


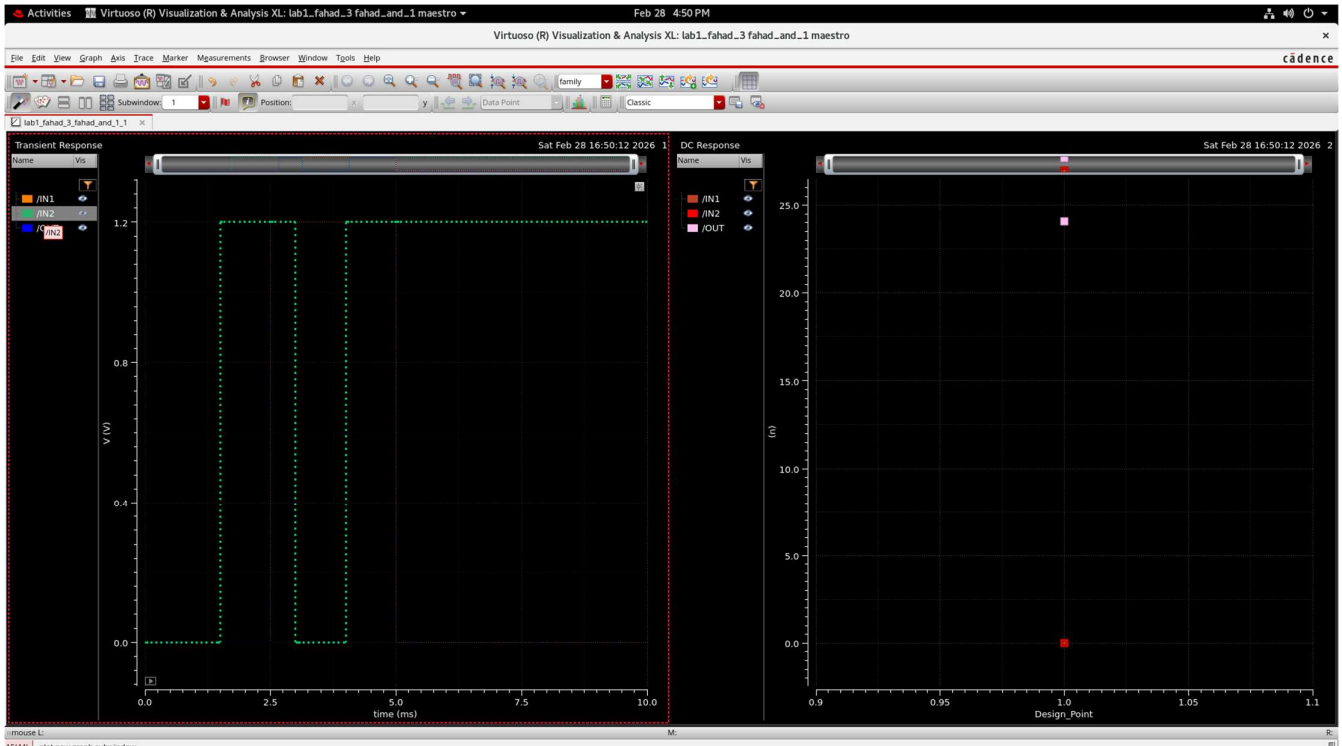
Exercise 2: Draw a circuit for AND gate using tsmc65N nch_mac and pch_mac transistor and observe its output.

- AND Gate



- Different Input Output relations




Activities
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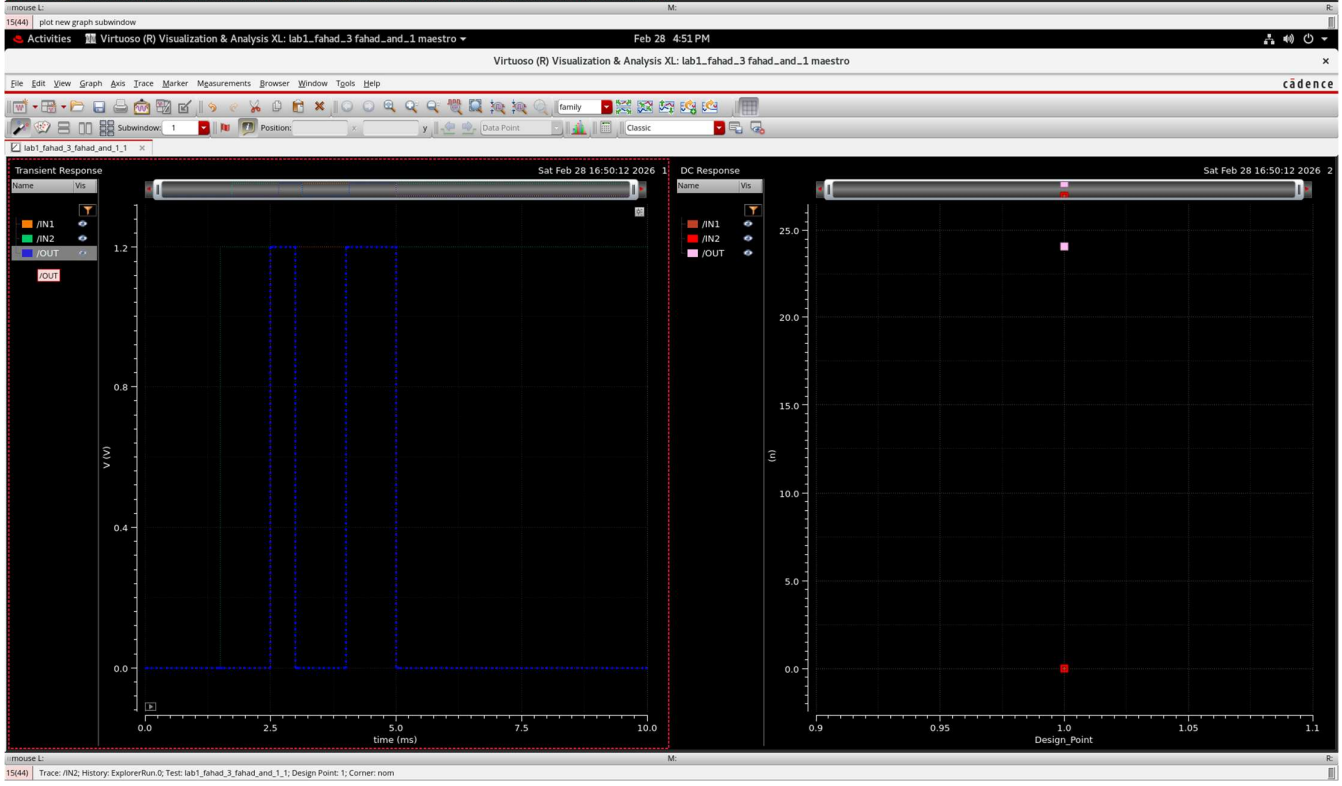
File Edit View Graph Axis Trace Marker Measurements Browser Window Tools Help
cadence
Transient Response
Sat Feb 28 16:50:12 2026
/IN1
/IN2
/OUT
V (V)
time (ms)
DC Response
Design_Point
plot new graph subwindow


Activities
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cadence
Transient Response
Sat Feb 28 16:50:12 2026
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/IN2
/OUT
V (V)
time (ms)
DC Response
Design_Point
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